EXPERIMENT NO : 6

TIME SERIES ANALYSIS OF WEBSITE TRAFFIC

AIM:

To time Series Analysis of Website Traffic.

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')

decomposition = seasonal_decompose(data['Page_Views'], model='additive')

decomposition.plot()

plt.show()

data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()

data[['Page_Views', 'Page_Views_MA']].plot()

plt.title('Page Views with Moving Average')

plt.show()

data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /
data['Page_Views'].std()

anomalies = data[data['Page_Views_Z'].abs() > 3]

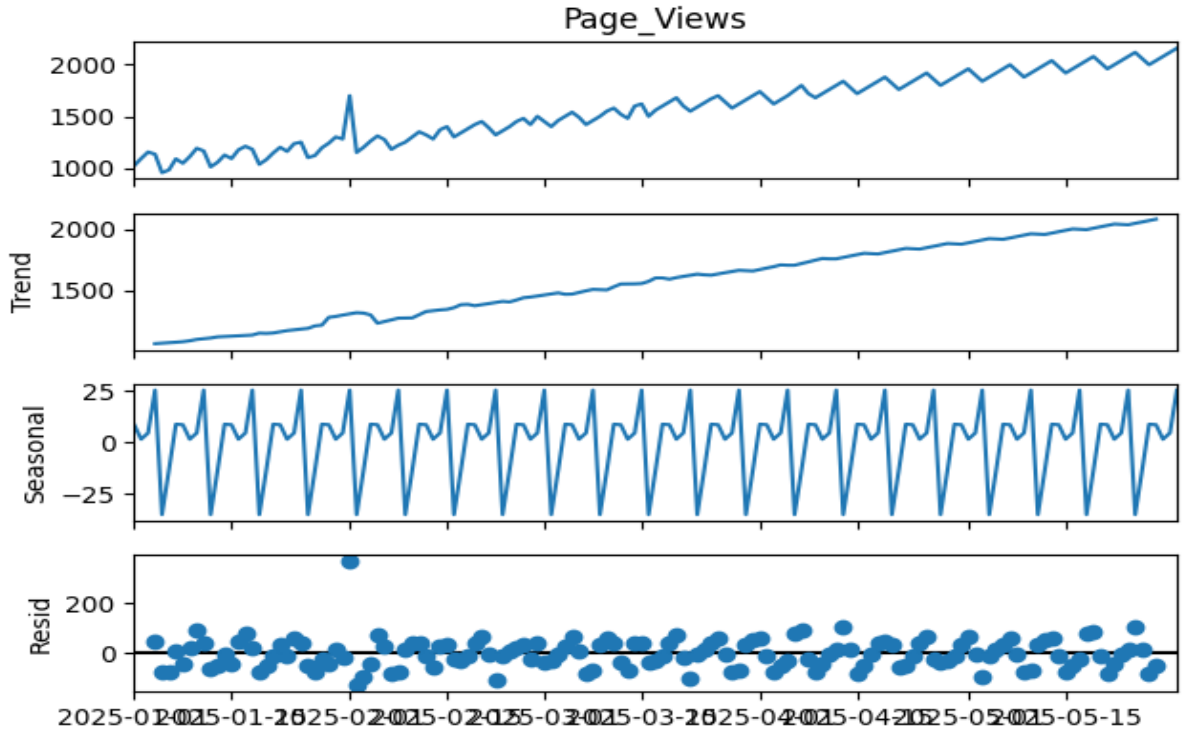
plt.plot(data.index, data['Page_Views'], label='Page Views')

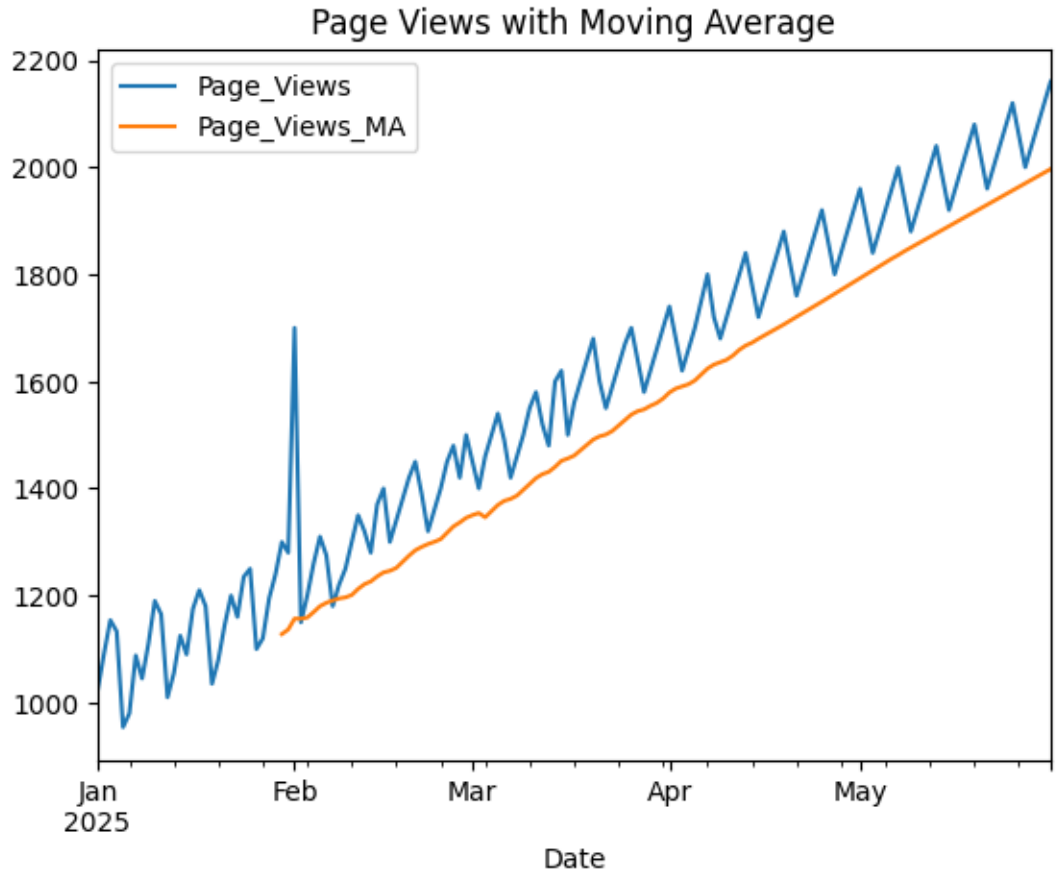
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')

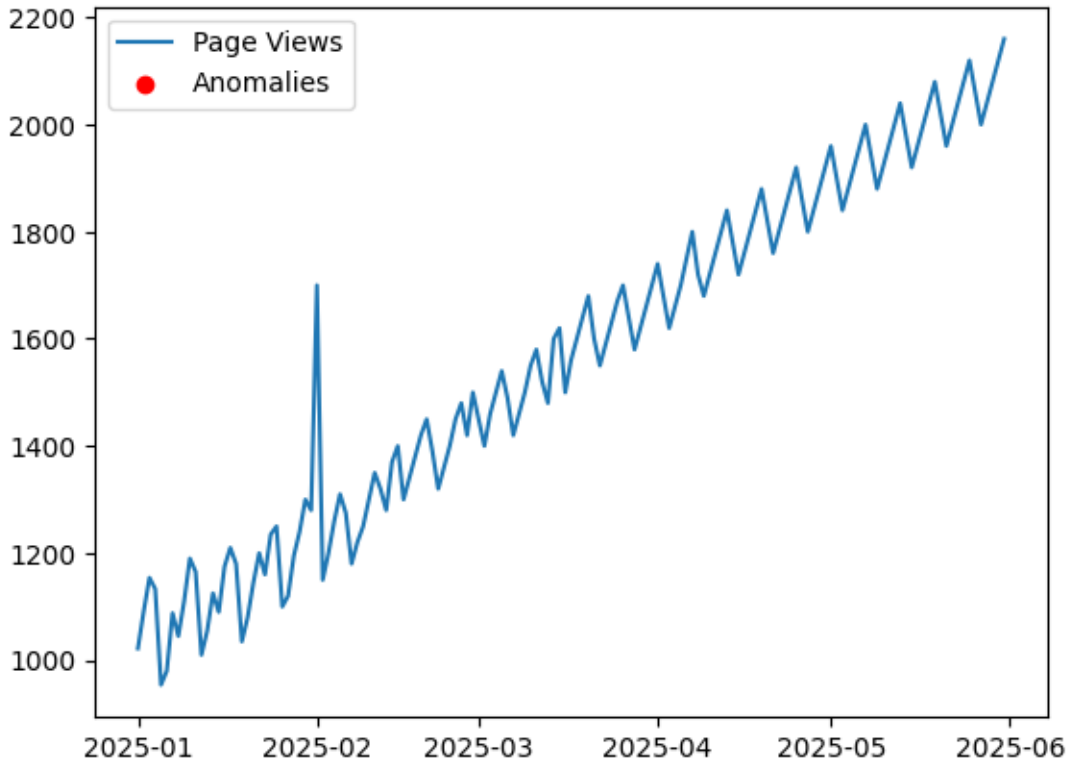
plt.legend()

plt.show()
```

OUTPUT:


Page_Views
2000
1500
1000
Trend
2000
1500
Seasonal
25
0
−25
Resid
200
0
2025-01-01
2025-01-15
2025-02-01
2025-02-15
2025-03-01
2025-03-15
2025-04-01
2025-04-15
2025-05-01
2025-05-15


Page Views with Moving Average
Page_Views
Page_Views_MA
2200
2000
1800
1600
1400
1200
1000
Jan
2025
Feb
Mar
Apr
May
Date



RESULT:

Thus, the given data was analysed successfully and required output was obtained.